Fluids and Lubricants Specifications

Battery Energy Storage System (BESS)
mtu EnergyPack QL
Version 2.0

A001069/01E

A Rolls-Royce solution

Table of Contents

DCL-ID: 0000059138 - 002

1 Preface

1.1 Preface

These Fluids and Lubricants Specifications apply to Battery Energy Storage Systems (BESS) from the manufacturer Rolls-Royce Solutions. Coolants for the converter cooler are specified.

Up-to-dateness of this document

The Fluids and Lubricants Specifications are revised or supplemented as required. Be sure to consult the latest version.

The most recent version can be retrieved from: http://www.mtu-solutions.com. Your contact will be happy to help you with any inquiries.

Safety data sheets for coolants

Safety data sheets for the various coolants may be obtained from the manufacturers concerned or from Rolls-Royce Solutions.

Symbols and presentation form used

The following instructions are highlighted in the text and must be observed:

Important
This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent material damage or destruction.

Note:

A note provides special instructions that must be followed when performing a task.

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2 Coolants

2.1 Coolants – General information

Coolant definition

Coolant	= coolant additive (concentrate) + freshwater to predefined mixing ratio ready for use in the converter cooler.

The corrosion-inhibiting effect of coolant is only ensured with the coolant circuit fully filled.

Coolants must be prepared from suitable freshwater and a coolant additive. For coolant additives see (→ Page 7).

Coolant must be prepared outside the Battery Energy Storage System.

Important

Mixing of different coolant additives and supplementary additives is prohibited.
Flushing with suitable freshwater is required when changing to a different coolant product.

Preventing damage to cooling system

- When topping up (following loss of coolant), ensure that not only water but also concentrate is added. The specified antifreeze and/or corrosion inhibitor concentration must be maintained.
- Use antifreeze mixed to 35% +/- 1% by volume (antifreeze protection down to -20 °C).
- Coolant circuits can not be completely drained as a rule. This means that residual quantities of old coolant or freshwater from the flushing process remain in the circuit. These residual quantities may dilute the coolant being filled. Check the coolant concentration in the coolant circuit and adjust as necessary.

Important

To ensure corrosion protection, the converter cooler should never be operated with pure water without adding a recommended anti-corrosion and antifreeze agent.

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